

Multigenerational Impacts of Early-Life Exposure to Medicaid

Early Life Exposure to Medicaid and the Safety Net

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Outline

- 1 Motivation
- 2 Policy Setting & Variation
- 3 Empirical design
- 4 First Stage and Validation
- 5 Main Results
- 6 Interpretation
- 7 Conclusion

Why Look Beyond the Treated Generation?

- Prenatal health shocks and investments have long-run effects
- Standard program evaluation usually stops at the directly treated cohort
- If effects transmit to children, policy benefits are understated

Research Question

- Does **in utero access to Medicaid** for women in the late 1970s/1980s improve **birth outcomes of their children**?
- Focus on outcomes that proxy early-life health and predict later outcomes:
 - Birth weight, low / very low birth weight
 - Gestational length, very preterm
 - Small for gestational age

Preview of Findings

- Those directly exposed to the expansion saw higher probability of enrollment at birth
- First generation cohort experienced reductions in likelihood of being born low birth weight
- Second generation cohort saw higher birth weights leading to fewer cases of very low birth weight and small for gestational age
- Suggestive evidence of less incidence of low birth weight and very preterm births for second generation

Medicaid Evolution Overview

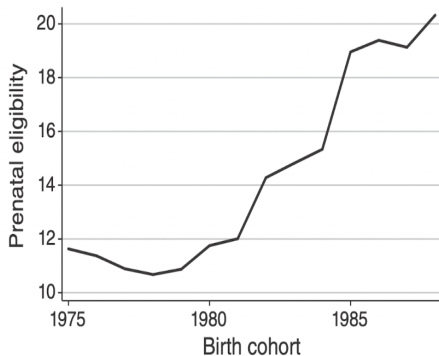
- Insurance supplement to AFDC implemented in 1965
- Strict eligibility criteria
 - Single women with one dependent child
 - Married women and women pregnant for the first time did not qualify
- AFDC financial eligibility varied widely across states, 24-99% of FPL (avg. 61%) in 1979

State Variation

- States could extend eligibility to first time pregnant women who would qualify for AFDC upon giving birth
- Eligibility could be offered to women who were not single mothers with dependent children (i.e. pregnant and married but qualified via income)
- In the early 80's federal mandate required Medicaid expansion to all women who met the financial eligibility criteria, even if family criteria was not met

Medicaid Coverage at Delivery Rises After Expansions

Panel A. Eligibility



Panel B. Coverage

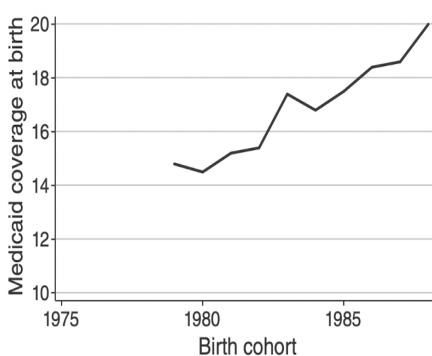


FIGURE 1. TRENDS IN PRENATAL MEDICAID ELIGIBILITY AND MEDICAID COVERAGE AT BIRTH, 1975 TO 1988

Targeted Prenatal Medicaid Expansions (1980–1985)

- States expanded eligibility for pregnant women at different times.
- Design uses **sharp eligibility jumps** (treated states) vs **no sharp jump** (controls).
- Key idea: compare cohorts born relative to each state's expansion year.

Treated vs Control States

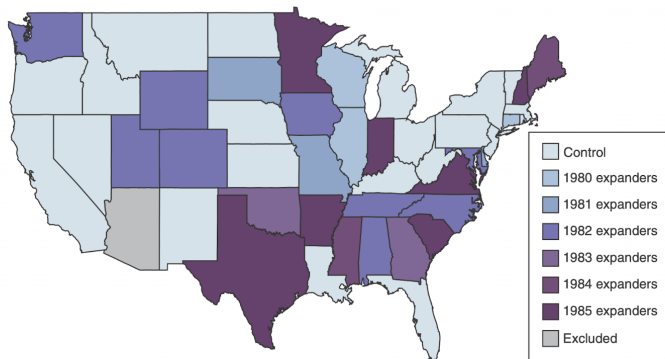


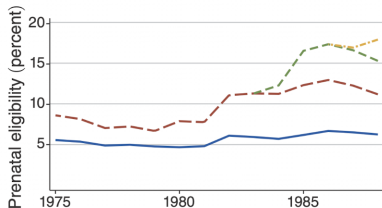
FIGURE 2. TREATMENT AND CONTROL STATES

Notes: See text for further details. Not pictured is Alaska (1982 expander) and Hawaii (control). Arizona is omitted due to the late start date of their Medicaid program.

- Treated states: first large discrete eligibility jump in 1980–1985.
- Controls: no large discrete jump in the same window.

Eligibility Trends

Panel A. Treated states



Panel B. Control states

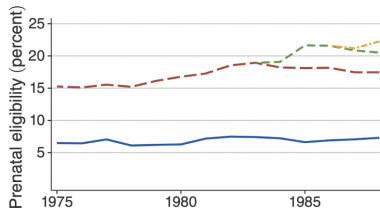


FIGURE 3. PRENATAL ELIGIBILITY BY SOURCE AND TREATMENT STATUS, 1975–1988

Notes: Authors' calculation from the CPS and Medicaid eligibility rules. Each line represents Medicaid eligibility through each of the state options and federal mandates. Specifically, we construct this figure by calculating the percent of women who would be eligible under each pathway in an additive fashion that reflects the order of eligibility pathways from lowest to highest pathway on the figure. These pathways are discussed in more detail in online Appendix A. All estimates are reported in percents.

Sources of eligibility growth: AFDC rules, targeted state options, targeted federal mandates, later income-based expansions.

Event-Study Design

For cohorts born in state n and year b :

$$y_{nb} = \alpha + \sum_{t \in [-5, 4] \setminus \{-1\}} \beta_t \mathbf{1}\{b - e_n = t\} + \eta_n + \lambda_b + \gamma X_{nb} + \varepsilon_{nb}.$$

- e_n : first year of the state's large eligibility jump.
- Omit $t = -1$ (year just before expansion): coefficients are relative to that baseline.
- Detrend using pre-period linear trend (state-specific), then estimate event study.

Linking Treatment Across Generations

- **First generation:** outcomes of infants born 1975–1988; treatment assigned by *state & year of birth*.
- **Second generation:** infants born to first-generation mothers; treatment assigned by *mother's state & year of birth*.
- Vital Statistics birth records provide:
 - infant outcomes
 - mother characteristics including state of birth and (approx.) birth year.

Data

- **Vital Statistics natality data (1975–2017):** universe of US births.
- **CPS ASEC + eligibility rules:** simulated prenatal eligibility.
- **NHDS hospital discharge data:** Medicaid coverage at delivery (first-stage validation).
 - Accessed restricted variables via FRDC, like state of birth
- **International Classification of Diseases:** labor and delivery hospitalization records (1979-1988)

Sample

- First-Generation:
 - ① Born 1975-1988 (observe full cohort fertility through age 28)
- Second-Generation
 - ① First born to mother in a treated cohort
 - ② Mother born in US (except Arizona)
 - ③ Born between 1990-2017

Methods: Mechanisms and Assumptions

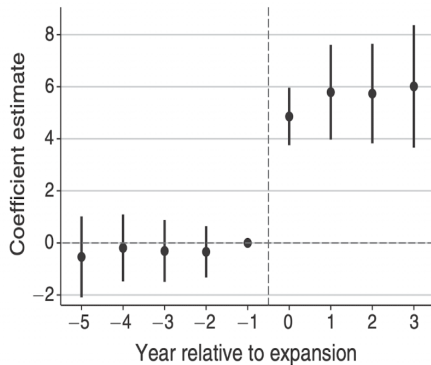
- **Event time:** relative to each treated state's *first large discrete* eligibility jump.
- **Detrending:** remove state-specific linear pretrends (estimated on the preperiod).
- **Controls:** state-year covariates (economy, welfare policy, abortion policy) + demographic controls.
- **Weighting:** by cohort size (population-weighted estimates).
- **Inference:** clustered SEs at the state level (state of birth / mother's state of birth).

Threats to Identification and Overcoming Them

- **Endogenous expansion timing:** pretrend diagnostics + detrending; robustness to alternative FE/control sets.
- **Other policy/economic changes:** state-year controls; robustness checks (e.g., add region-year FE, drop control states).
- **Composition (fertility/migration) changes:** sensitivity to weighting and sample restrictions.
- **Confounding from later childhood Medicaid expansions:** add later-eligibility measures; patterns remain similar.
- **Treatment assignment error (2nd gen):** use mother's state/year of birth; stability across specs.

Actual vs. Simulated Eligibility

Panel A. Actual eligibility



Panel B. Simulated eligibility

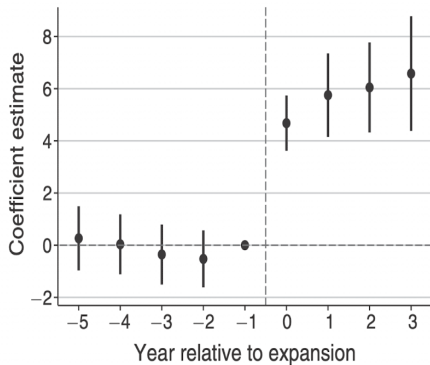


FIGURE 5. EVENT STUDY COEFFICIENTS FOR PRENATAL ELIGIBILITY

First-Generation: Low Birth Weight Declines

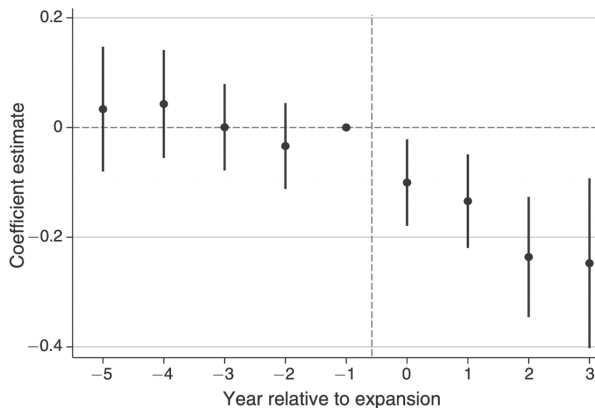
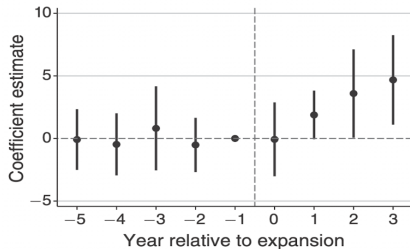


FIGURE 7. EVENT STUDY FOR FIRST GENERATION OUTCOME: LOW BIRTH WEIGHT

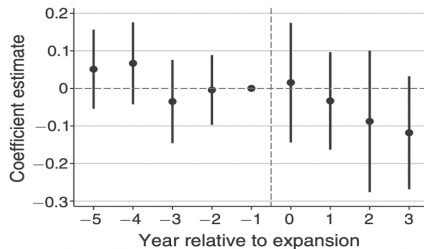
Notes: Coefficient estimates are reported in percentage points. Estimated for infants born in 1975–1988. Preperiod trend is estimated and removed from all observations for each state prior to the event study estimation. For treated states, this is estimated using all preperiod years for each state. For control states, we use the period 1975–1981 to estimate this trend. Regressions are weighted by birth cohort size and include state of birth and year of birth fixed effects and controls for state-year variables (unemployment rate, personal income per capita, maximum welfare benefit for a family of four, indicators for state parental consent and notification laws and state Medicaid restric-

Second-Generation: Birth Weight Improves

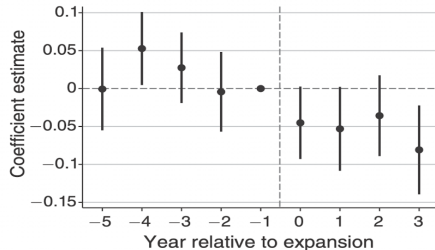
Panel A. Birthweight



Panel B. Low birthweight

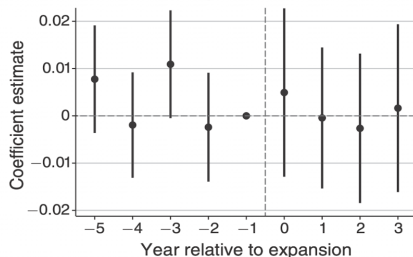


Panel C. Very low birthweight

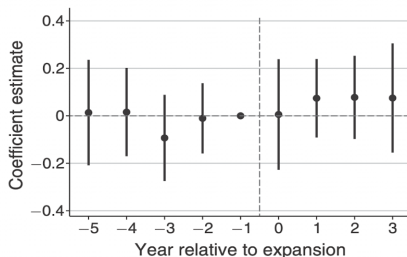


Second-Generation: Gestational Outcomes

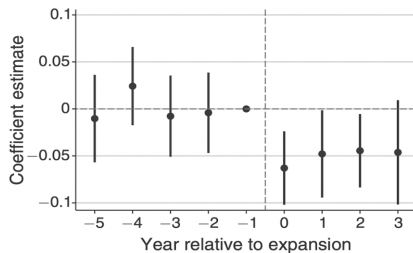
Panel A. Gestational length



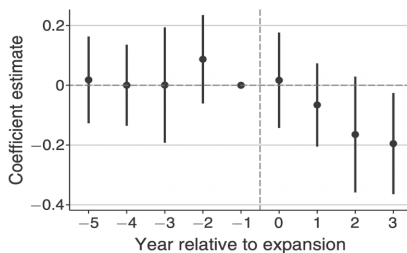
Panel B. Preterm



Panel C. Very preterm

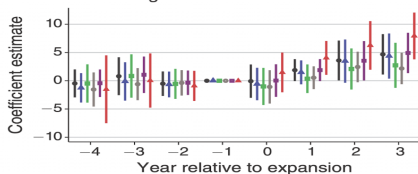


Panel D. Small for gestational age

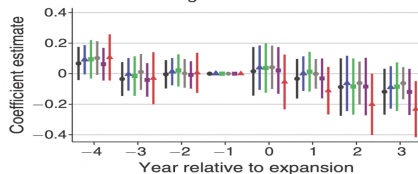


Robustness: Alternative Controls and Specifications

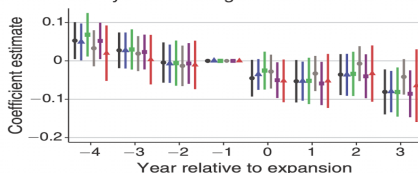
Panel A. Birthweight



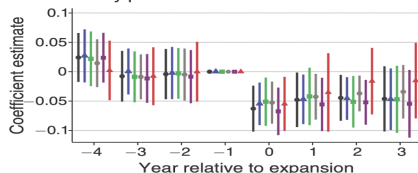
Panel B. Low birthweight



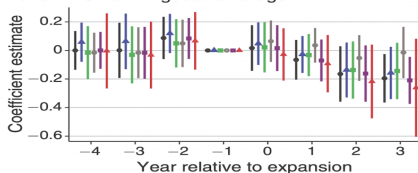
Panel C. Very low birthweight



Panel D. Very preterm



Panel E. Small for gestational age



- Baseline
- Control region* year FE
- Mother cohort weights
- ▲ No state* year controls
- Don't detrend
- ▲ No control states

Mechanisms

- Biological pathway: improved maternal health endowments at birth
- Maternal health during pregnancy: lower chronic disease risk, better prenatal environment
- Non-biological channels: stress reduction, resources, and prenatal care

Policy Interpretation

- Multigenerational benefits imply standard program evaluations may understate returns
- Medicaid expansions are not only safety-net insurance:
 - they change early-life health trajectories, which lead to improved later-life circumstances
 - and those changes can transmit to the next generation

Takeaways

- Prenatal Medicaid expansions improved first-generation infant health, in line with past results
- Those improvements transmit: second-generation birth outcomes also improve, though to a lesser degree
- Implementation of policy should keep the potential spillovers in mind, as the initial payoff could essentially be thought of as investment into future outcomes